

Fig. 5: Path of automated line-following robot

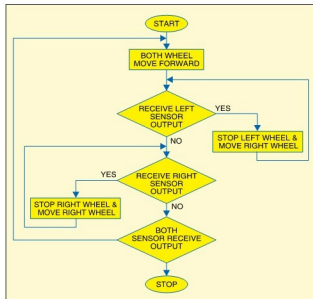


Fig. 6: Program flow-chart for automated line-following robot

microcontroller having 4 kB of Flash, 128 bytes of RAM, 32 I/O lines, two 16-bit timers/counters, a five-vector two-level interrupt architecture, on-chip oscillator and clock circuitry. A 12MHz crystal is used for providing the basic clock frequency. All I/O pins are reset to '1' as soon as RST pin goes high. Holding RST pin high for two machine cycles while the oscillator is running resets the device. Power-on reset is derived from resistor R5 and

capacitor C1. Switch S2 is used for manual reset. The microcontroller, based on the inputs from sensor T1 (say, left) and sensor T2 (say, right), controls the motor to make the robot turn left, turn right or move forward.

Port pins P2.0, P2.1, P2.2 and P2.3 are connected to pins 15, 10, 7 and 2 of motor driver L293D. Port pins P2.0 and P2.1 are used for controlling the right motor, while port pins P2.2 and P2.3 are used for controlling the left motor. Three wheels can be used for this robot—one on the front and two at the rear. Front wheel can rotate in any direction as specified by the rear wheel. To make the robot turn left, the left-side motor should stop and the right-side

motor should rotate in the clockwise direction. Similarly, to make the robot turn right, the right-side motor should stop and the left-side motor should rotate in clockwise direction. For forward motion, both the motors should rotate in clockwise direction.

Working

An actual-size, single-side PCB for the automated line-following robot is shown in Fig. 3 and its component

layout in Fig. 4. Fig. 5 shows the path of the line-follower robot, where 'L' is the left sensor and 'R' is the right sensor.

At the start, when the robot is at point 'A,' sensors T1 and T2 are above the black surface and port pins P3.0 and P3.1 of the microcontroller receive logic '0'. As a result, the robot moves forward in straight direction.

At point 'B,' a left turn is encountered, and the left sensor comes above the white surface, whereas the right sensor remains above the black surface. Port pin P3.0 of the microcontroller receives logic '1' from the left sensor and port pin P3.1 receives logic '0' from the right sensor. As a result, the left motor stops and the right motor rotates, to make the robot turn left. This process continues until the left sensor comes above the black background.

Similarly, at point 'C,' where a right turn is encountered, the same procedure for right turn is executed. When both the sensors are at the white surface, the robot should stop. The output of the microcontroller (IC2) depends on the inputs received at its port pins P3.0 and P3.1 as shown in table.

Software

The source program for the project is written in Assembly language and assembled using Metalink's ASM51 assembler, which is freely available on the Internet for download. It is well commented for easy understanding and works as per the flow-chart shown in Fig. 6. The hex file 'robot.hex' is to be burnt into the microcontroller.

EFY note. The source code and other relevant files of this article are included in this month's EFY-CD.

ROBOT.ASM

```

SMOD51
    ORG 0000H
    LMP MAIN
    ORG 0030H
MAIN: SETB P3.0 ;input for left sensor
    SETB P3.1 ;input for right sensor
    SJMP AGAIN
AGAIN: JB P3.0,NEXT
    JB P3.1,GO
    CLR P2.0

```

```

    SETB P2.1
    CLR P2.2
    SETB P2.3
    SJMP AGAIN
GO:   CLR P2.0
    SETB P2.1
    CLR P2.2
    CLR P2.3
    SJMP AGAIN
NEXT: JB P3.1,GO1
    CLR P2.0
    CLR P2.1

```

```

    CLR P2.2
    SETB P2.3
    SJMP AGAIN
    CLR P2.0
    CLR P2.1
    CLR P2.2
    CLR P2.3
    SJMP AGAIN
    SJMP HERE
END

```